HAT-P-54b

R-1625

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-54_190108` · created 2026-08-02T04:59:58+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458491.8172 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.199 +/- 0.053
Transit depth (Rp/Rs)^2	3.97 +/- 2.13 [%]
Orbital Inclination (inc)	83.5 +/- 2.31
Airmass coefficient 1 (a1)	1.497 +/- 0.069
Airmass coefficient 2 (a2)	-0.38 +/- 0.11
Scatter in the residuals of the lightcurve fit is	4.52 %
Best Comparison Star	#1 - [522, 172]
Optimal Method	PSF photometry
Transit Duration (day)	0.041 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.199 ± 0.053 vs 0.1572 (deviation 0.79σ)
Duration fit vs published	0.041 d vs 0.075 d (deviation 1.36σ)
Mid-time O–C	-4.2 min
Depth SNR	3.8
Residual scatter	4.52 %

Light curve

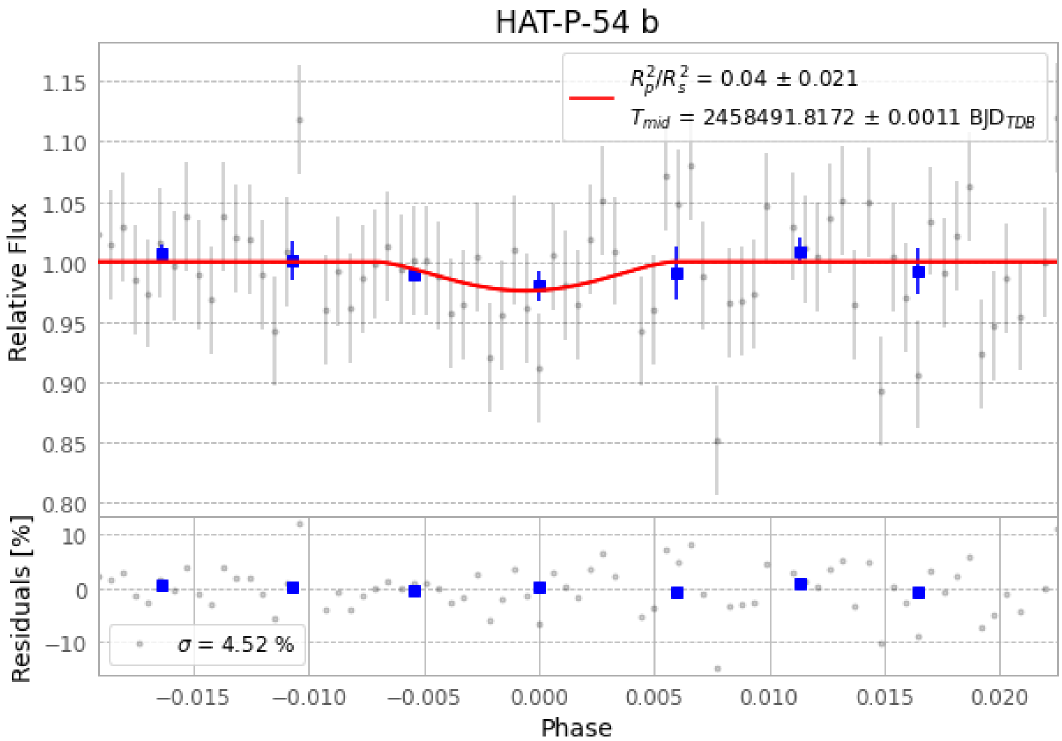

Figure 1 — FinalLightCurve_HAT-P-54 b_2019-01-07.png (SHA-256 b8b42493617aa0e0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 263], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6ca665c70580b351…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `HATP-541190108054212.FITS` → `HATP-541190108093012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-190108.log (fe75de1127fc…), FinalLightCurve_HAT-P-54 b_2019-01-07.png (b8b42493617a…), FinalLightCurve_HAT-P-54 b_2019-01-07.csv (e9d1da3db78b…)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 04:59Z.